

Transformer Architecture

The Transformer was introduced in 'Attention is All You Need' (2017).

It replaced recurrent networks with a pure attention mechanism, enabling parallelism during training and yielding state-of-the-art results on NLP benchmarks. Self-attention allows the model to relate tokens to each other regardless of their distance in the sequence.

Self-Attention and Multi-Head Attention

Self-attention computes $\text{Attention}(Q, K, V) = \text{softmax}(QK^T/\sqrt{d_k})V$.

Multi-head attention runs h attention functions in parallel across different learned subspaces, concatenates the results, and projects.

This allows the model to attend to information from different representation subspaces simultaneously.

Applications and Variants

Encoder-only models (BERT) are used for classification tasks.

Decoder-only models (GPT) are used for language generation.

Encoder-decoder models handle sequence-to-sequence tasks.

Transformers power modern LLMs, vision models, and code assistants.

Fine-tuning on downstream tasks achieves strong performance.